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CHM 221 — Past Questions & Materials
by Renance

Course: CHM 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200

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Paper 1 of 5 — CHM 221 Analytical chemistry 2023/2024

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATION 2023/2024 SESSION OCTOBER 2024
COURSE TITLE: ANALYTICAL CHEMISTRY 1 COURS CODE: CHM221
INSTRUCTION: ANSWER THREE QUESTIONS ONLY TIME: 2HOURS

QUESTION ONE

- (a) Describe the steps involved in the process of solvent extraction. **8marks**
- (b) Compare and contrast, Gas Chromatography (GC) and Paper Chromatography (PC), highlighting their principles and applications. **10marks**
- (c) Define ion exchange and explain the principle behind this separation method. **5marks**

QUESTION TWO

- (a) Discuss two industrial applications of ion exchange in water purification or biomolecule separation. **8marks**
- (b) What are the key factors that influence the separation of compounds in column chromatography? **5marks**
- (c) Explain the principles of electrophoresis and how it can be used for the separation of proteins and nucleic acid. **10marks**

QUESTION THREE

- (a) Distinguish between accuracy and precision **5mks**
- (b) A standard containing 50.0 ppm Cl was analysed by two methods, 1 and 2. The results from method 1 averaged at 50.06 ppm Cl while that from method 2 averaged at 50.38ppm Cl. Explain the conclusion that can be drawn from the results of the two methods. **8marks**
- (c) The following results were obtained in the gravimetric determination of a precipitate, **10marks**

- i. 22.64
- ii. 22.54
- iii. 22.22
- iv. 22.69

Apply a statistical test to determine if all the four values can be used to evaluate the mean or not, and determine the mean. Use the table below.

(n)	Value of R_1/R_2 at 99% probability level
4	9.0
6	3.3
10	2.1
(n) = number of readings	

2

QUESTION FOUR

- (a) Give three possible causes of impurities in a precipitate, and suggest three ways each of the impurities could be removed.
- (b) Carbon and hydrogen content in a sample were determined by combustion analysis. When a 0.5123g sample was carried out through the analysis, 1.2119 of CO_2 and 0.2482g of H_2O were collected.
- What are the percentage w/w of carbon and hydrogen in the compound?
 - What is the empirical formula of the compound?
- (c) The calcium in a 200.0mL sample of a natural water was determined by precipitating the cation as calcium oxalate (CaC_2O_4). The precipitate was filtered, washed and ignited in a crucible with an empty mass of 26.6002g. The mass of the crucible plus CaO (56.077g/mole) was 26.7134g. Calculate the concentration of calcium (40.078g/mol) in water in unit of grams per 100mL of the water.

QUESTION FIVE

- (a) What is neutralisation titration?
- (b) Calculate the mass of silver carbonate (Ag_2CO_3) that is formed when 25.00 of 0.20mol dm^{-3} AgNO_3 solution is made to react with 50.00cm^3 of 0.08mol dm^{-3} sodium carbonate (Na_2CO_3) solution.
- (c) A 25.00cm^3 portion of 0.30mol dm^{-3} of sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$) solution was warmed and titrated against acidified potassium tetraoxomanganate(vii) solution. If 45.00cm^3 of KMnO_4 solution was needed for the redox titration, calculate the concentration of KMnO_4 solution in i. mol dm^{-3} ii. g dm^{-3}

END OF PAPER

Paper 2 of 5 — CHM 221 Past Questions - Analytical Chemistry (2018/2019)

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATION, 2018/2019 SESSION

ANALYTICAL CHEMISTRY, TIME: 2HOURS

INSTRUCTION: ANSWER Questions ONLY

(a) If precision tends to blur the existence of systematic errors, explain two different auxiliary measurements that could be performed to check them.

(b) Two different determinations on a sample of rock using different methods showed that the percentage magnesium content of the rock sample is as stated below:

Method 1 — 10.05 ± 0.35

Method 2 : 10.45 ± 0.05,

4 a

5 Pett

Which of the results is more precise and, why?

D Tapert

Which is more accurate and why? Explain;

(g) What is the relationship between bias and systematic error?

(a) Name the steps involved in calculating the standard deviation of a given set of analytical data.

(b) A geochemist analysed the levels of lead metals in a sample of crude oil from Bonga Oil Well in the

Niger, Delta sedimentary basin using atomic absorption spectrometer and inductively coupled plasma

spectrometer, and their means came out to be 22.54 ppm and 22.4 ppm. In both instruments the standard deviation of the data was calculated to be 0.08. Using t — test statistics, test the significance of

the results at 99% probability level. $t_{0.01, 19} = 2.861$

(c) State two other statistical tests that can be used to test for the level of significance of a set of analytical data.

3 (a) Explain the sources of precipitation in gravimetry.

Derwentlee "fig Beet : 1 Oy

(b) Discuss the approaches used in causing a colloid to coagulate

(c) How long will it take to remove all copper from one litre of 1.0 M solution of Cu^{2+}

assuming a current

of 0.2 Amperes is applied and the electrolytic cell achieved a 50% efficiency?

ity

4 (a) Disti

fish j in strict terms, between' equivalence point and end-point,

(ey (i i) What i isa primary standard? A standard 'solution?

a plain the desired Piohe tes of a primary standard.

tii) What are the qualities ofa 'primary standard solution?

(c) Differentiate between molarity and normality.

me Jae END OF PAPER

Paper 3 of 5 — CHM221, 2024 ANALYTICAL CHEMISTRY past question

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATION 2023/2024 SESSION OCTOBER 2024
COURSE TITLE: ANALYTICAL CHEMISTRY I COURS CODE: CHM221
INSTRUCTION: ANSWER THREE QUESTIONS ONLY TIME: 2HOURS

QUESTION ONE

- Describe the steps involved in the process of solvent extraction. **8marks**
- Compare and contrast, Gas Chromatography (GC) and Paper Chromatography (PC), highlighting their principles and applications. **10marks**
- Define ion exchange and explain the principle behind this separation method. **5marks**

QUESTION TWO

- Discuss two industrial applications of ion exchange in water purification or biomolecule separation. **8marks**
- What are the key factors that influence the separation of compounds in column chromatography? **5marks**
- Explain the principles of electrophoresis and how it can be used for the separation of proteins and nucleic acid. **10marks**

QUESTION THREE

- Distinguish between accuracy and precision **5marks**
- A standard containing 50.0 ppm Cl was analysed by two methods, 1 and 2. The results from method 1 averaged at 50.06 ppm Cl while that from method 2 averaged at 50.38ppm Cl. Explain the conclusion that can be drawn from the results of the two methods. **8marks**
- The following results were obtained in the gravimetric determination of a precipitate. **10marks**
 - 22.64
 - 22.54
 - 22.22
 - 22.69

Apply a statistical test to determine if all the four values can be used to evaluate the mean or not, and determine the mean. Use the table below.

(n)	Value of R_1/R_2 at 99% probability level
4	9.0
6	3.3
10	2.1

(n) = number of readings

Paper 4 of 5 — CHM 221 LATTICE ENERGY

Level 200 • First Semester • Handout

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LATTICE ENERGY

The Born-Haber Cycle

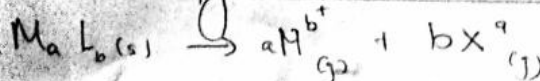
Ionic solids tend to be very stable compounds. The enthalpies of formation of the ionic molecules cannot alone account for this stability. These compounds have an additional stability due to the lattice energy of the solid structure. However, lattice energy cannot be directly measured. The Born-Haber cycle allows us to understand and determine the lattice energies of ionic solids.

Lattice energy is a measure of the bond strength in an ionic compound. It is defined as the energy required to separate a mole of an ionic solid into gaseous ions. Lattice energy can not be measured empirically but it can be calculated using electrostatics or estimated using the Born-Haber cycle.

Lattice energy can also be defined as the heat of formation for ions of opposite charge in the gas phase to combine into an ionic solid.

Lattice Energy however, can also be said to be a potential energy. Although there are two commonly used ways of defining lattice enthalpy which directly contradict each other, and we will primarily

- a) The Lattice energy is the energy required to break apart an ionic solid and convert its component atoms into gaseous ions. This definition causes the value of the Lattice energy to always be positive. Since this will always be an endothermic reaction. In other words, you are looking at a downward arrow in the figure below.



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Question: _____

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- (b) The lattice energy is the energy released when gaseous ions bond to form an ionic solid. As implied in the definition, this process will always be exothermic, and thus the value for Lattice energy will be negative. Its values are usually expressed with the units kJ/mol . In other words, you are looking at an upward arrow on the figure below.

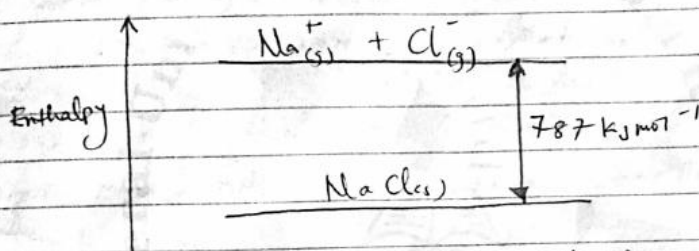
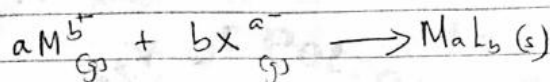


Fig. 1: Enthalpy diagram for $\text{NaCl}(s)$. The solid is more stable than the gaseous ions by 787 kJ mol^{-1} .

Values of lattice energies for various solids have been given below especially for some common solids

Lattice Energies (kJ/mol) of some salts

LiF	1036	KI	649
LiCl	853	MgF_2	2957
LiBr	807	MgCl_2	2526
LiI	757	MgBr_2	2440
NaF	923	MgI_2	2327
NaCl	786		
NaBr	747		
NaI	704		
KF	821		
KCl	715		
KBr	682		

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Factors Affecting Lattice Enthalpy

Lattice enthalpy is a measure of the strength of the forces between the ions in an ionic solid. These forces are only completely broken when the ions are present as gaseous ions, scattered so far apart that there is negligible attraction between them. The greater the lattice enthalpy, the stronger the forces. The force of attraction between oppositely charged particles is described by Coulomb's law. Thus

- a) As the ionic radii of either the cation or anion increases, the lattice energies decrease.
- b) The solids consisting of divalent ions have much larger lattice energies than solids with monovalent ions.

The factors affecting lattice enthalpy are +

- a) The charges on the ions
- b) The radii of the ions

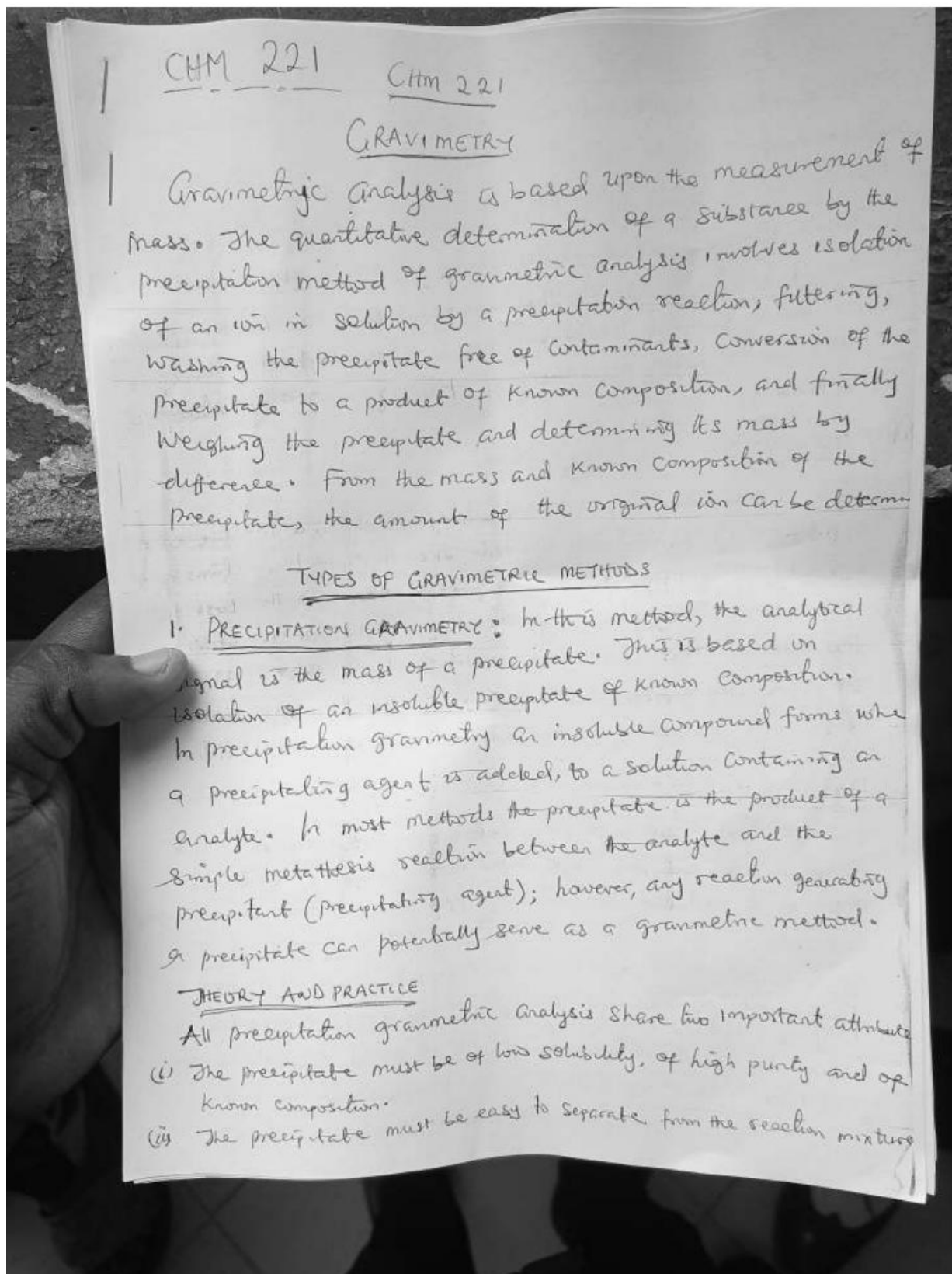
Excellence and Reliability

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Paper 5 of 5 — CHM 221 ANALYTICS CHEMISTRY

Level 200 • First Semester • Handout

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Evaluating Precipitation Gravimetry

To evaluate precipitation gravimetry, the following characteristics are required:

(i) Scale of operation: The scale of operation for precipitation gravimetry is limited by the sensitivity of the balance and the availability of sample. The analysis of trace level analytes or micro samples usually requires a microanalytical balance, that can measure an accuracy of $\pm 0.1\%$ with a sensitivity of $\pm 0.1\text{mg}$.

(ii) Accuracy: For a mass sample, containing a major analyte, a relative error of 0.1–0.2% is routinely achieved. The principle limitations are solubility losses, impurities in the precipitate, and the loss of precipitate during handling. When it is difficult to obtain a precipitate that is free from impurities, it may be possible to determine an empirical relationship between the precipitate's mass and the mass of the analyte by an appropriate dilution.

(iii) Precision: The relative precision of precipitation gravimetry depends on the sample size and the precipitates mass. Larger precipitates require a relative precision of ppm while smaller ones require 1–2 ppt.

(iv) Sensitivity: In precipitation gravimetry, the following general equation can be written by relating the signal (grams of precipitate) to the absolute amount of analyte in the sample.

$$\text{grams precipitate} = K \times \text{grams analyte}$$

Where K , the method's sensitivity, is determined by the stoichiometry between the precipitate and the analyte. Consider, for example, the determination of Fe_2O_3 . Using conservation of mass for iron, the precipitate mass is

$$g \text{ Fe}_2\text{O}_3 = g \text{ Fe} \times \frac{1 \text{ mol Fe}}{\text{Aw Fe}} \times \frac{\text{FW Fe}_2\text{O}_3}{2 \text{ mol Fe}}$$

and the value of K is

$$K = \frac{1}{2} \times \frac{\text{FW Fe}_2\text{O}_3}{\text{Aw Fe}}$$

Ways to improve a method's sensitivity

- To increase the ratio of the precipitate's molar mass to that of the analyte; this helps to form a precipitate with the largest possible formula weight.
- By forming a precipitate that contains fewer units of the analyte.

(ii) Selectivity:

Due to the chemical nature of the precipitation process, precipitates are usually not selective for a single analyte. For example, Ag^+ is not a selective precipitant for chloride, because it also forms precipitates with bromide and iodide. Interferents are often a serious problem and must be considered if accurate results are to be obtained.

(iii) Time, Cost and Equipment

Precipitate gravimetry is time intensive and rarely practical if you have a large number of samples to analyze. Equipment needed are few—beakers, filtering devices, ovens or burners and balances, routinely available in most laboratories and easy to maintain.

VOLATILIZATION GRAVIMETRY

2

This second method of gravimetry is to thermally or chemically decompose the sample and measure the resulting change in mass. Alternatively, we can trap and weigh a volatile decomposition product.

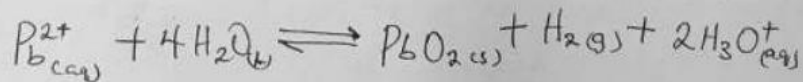
THEORY AND PRACTICE OF VOLATILIZATION GRAVIMETRY

Whether an analysis is direct or indirect, volatilization gravimetry usually requires that we know the products of the decomposition reaction. This is rarely a problem for organic compounds, which typically decompose to form single gases such as CO_2 , H_2O or N_2 . For ~~an~~ inorganic compounds, however, the products often depend on the decomposition temperature.

For example, in determining the moisture content of bread, we use thermal energy to vaporize the water ^{in the} sample. To determine the amount of carbon in an organic compound, we use the chemical energy of combustion to convert it to CO_2 .

3. ELECTROGRAVIMETRY

In electrogravimetry, we deposit the analyte as a solid film (an electrode) in an electrochemical cell. For example, if we suspend a pair of Pt electrodes in the sample (analyte) containing Pb^{2+} and apply a sufficiently positive potential between them for a long time, we can obtain PbO_2 .



Oxidizing Pb^{2+} deposits PbO_2 on the Pt anode. If we weigh the anode before and after applying the potential, the change in its

mass gives the mass of PbO_2 and from the reaction's stoichiometry the amount of Pb^{2+} in the sample. Another example is the reduction of Cu^{2+} to Cu at a Pt cathode.

4. PARTICULATE GRAVIMETRY

In particulate gravimetry, the analyte is determined by separating it from the sample's matrix using a filtration or an extraction. The determination of total suspended solids is one example of particulate gravimetry. For example;

After collecting a sample, you pass it through a preweighed filter that retains the suspended solids, and dry the filter and solids to remove any residual moisture. The mass of suspended solid is the difference between the filter's final mass and its original mass.

THEORY AND PRACTICE OF PARTICULATE GRAVIMETRY

There are two methods for separating a particulate analyte from its matrix. The most common method is filtration, in which we separate solid particulates from their gas, liquid or solid matrix. A second method, which is useful for gas particles, solutes and solids is an extraction.

Assignment

1. Explain the two methods commonly used in particulate gravimetry
 - (a) Filtration
 - (b) Extraction

USES / IMPORTANCE OF GRAVIMETRIC ANALYSIS

1. In gravimetric analysis, if methods are followed carefully provides exceedingly precise analysis.
2. It is used to determine the atomic masses of many elements to six figure accuracy.
3. Gravimetry provides very little room for instrumental error and does not require a series of standards for calculation of and unknown.
4. The methods do not require expensive equipment.
5. It can be used to calibrate other instruments in lieu of reference standards if performed correctly.

Example 1

Chloride is precipitated as silver chloride, which is weighed on the weight of chloride calculated from the weight of silver chloride. If a given sample produced 1.000 g of silver chloride, what weight of chloride was present.

Solution



Since there is one atomic mass of Cl in each molecular mass of AgCl

$$\text{No of atomic mass of Cl} = \text{Nm of AgCl}$$

$$\text{No of atomic mass of Cl} = \frac{\text{g of Cl}}{\text{at wt Cl}} = \frac{35.45}{35.45}$$

$$\text{No of moles of AgCl} = \frac{\text{g of AgCl}}{\text{Nm of AgCl}} = \frac{1.000}{143.3}$$

$$\frac{\text{2g of Cl}}{35.45} = \frac{1.000}{143.3}$$

$$\text{2g Cl} = \frac{35.45 \times 1.000}{143.3} = 0.2474 \text{ g}$$

$$\% \text{ of Fe} = \frac{\text{Mass of ppt}}{\text{Mass of sample}} \times 100$$

$$\frac{0.0728}{0.4873} \times 100 = 15.02\%$$

Alternative method.

$$\% \text{ of analyte} = \frac{\text{wt of ppt} \times \text{g (gravimetric factor)} \times 100}{\text{wt of sample}}$$

$$\% \text{ of Fe} = \frac{\text{Fe}}{\text{FeSO}_4} = \frac{56}{160} = 0.85$$

$$\% \text{ of analyte} = \frac{0.2591 \times 0.35 \times 100}{0.4873} = 15.02\%$$

Example 4

The C and H in a second sample were determined by a combustion analysis. When a 0.5123 g sample was carried through the analysis, 1.2119 g of CO_2 and 0.2482 g of

H_2O were collected.

(i) What are the % Wt of C and H in the compound?

(ii) What is the compound's empirical formula?

Solution.

$$\text{(i) Mole} = \frac{\text{mass}}{\text{molar mass}}$$

$$\text{Mole of } \text{CO}_2 = \frac{1.2119 \text{ g}}{44 \text{ g/mol}} = 0.0275 \text{ mol}$$

$$\text{Therefore mass C} = \text{mole} \times \text{MM}$$

$$= 0.0275 \text{ mol} \times 12 \text{ g/mol}$$

$$= 0.33 \text{ g}$$

$$\% \text{ of C in sample} = \frac{\text{wt of analyte}}{\text{wt of sample}} \times 100$$



Calculate the percent SO_3 in a sample of gypsum if a sample produced a precipitate of BaSO_4 weighing 0.9875 g.

Solution

$$\% \text{SO}_3 = \frac{\text{Wt of precipitate} \times \text{factor} \times 100}{\text{Wt of sample}}$$

Gravimetric factor (GF) is used for simplifying gravimetric calculations.

$$\text{GF} = \frac{\text{Aw of analyte}}{\text{Fwt of ppt}} = \frac{\text{SO}_3}{\text{BaSO}_4} = \frac{80.06}{233.42}$$

$$\therefore \text{Factor} = 0.3430$$

$$\% \text{SO}_3 = \frac{0.9875 \times 0.3430 \times 100}{0.7560} = \underline{\underline{44.80\%}}$$

Example 3

The iron content of an organometallic compound was determined by treating a 0.4873 g sample with HNO_3 and heating to volatilize the organic material. After ignition, the residue of Fe_2O_3 was found to weigh 0.2091 g. What is the % w/w Fe in this compound?

Solution

$$\text{mole} = \frac{\text{Mass}}{\text{molar mass}}$$

$$\text{Mole Fe}_2\text{O}_3 = \frac{0.2091 \text{ g}}{160 \text{ g/mol}} = 0.001306 \text{ mol}$$

$$\therefore \text{mass of Fe} = \text{mole} \times \text{MM} = 0.001306 \text{ mol} \times 56 \text{ g/mol} = \underline{\underline{0.07289 \text{ g}}}$$

$\frac{0.33}{0.5123} \times 100 = \underline{64.42\%}$

(b) mole of $H_2O = \frac{0.2482 \text{ g}}{18 \text{ g/mol}} = 0.0138 \text{ mol}$

Therefore mass of H = mole $\times$ MM
 $= 0.0138 \times 2 \text{ (2 atoms of H}_2 \text{ in H}_2\text{O)}$
 $= 0.0276 \text{ g}$

$\% \text{ of H}_2 \text{ in Sample} = \frac{\text{wt of analyte}}{\text{wt of sample}} \times 100$
 $= \frac{0.0276 \text{ g}}{0.5123 \text{ g}} \times 100 = \underline{5.39\%}$

(ii) Empirical formula

$C = \frac{64.42}{12} = 5.37$

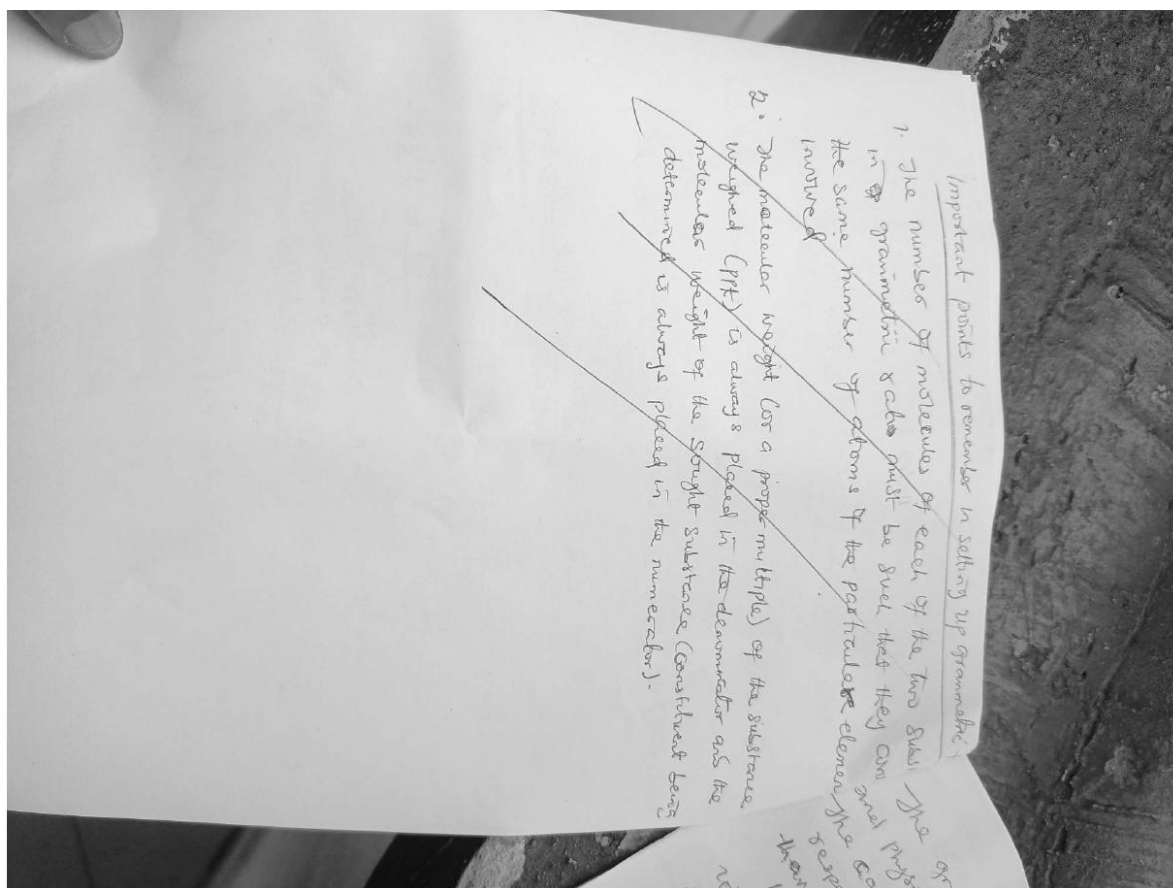
$H = \frac{5.39}{1} = 5.39$

Divide through by the smallest value

$C = \frac{5.37}{5.37} = 1.0$

$\frac{5.39}{5.37} = 1.003$

Therefore, the empirical formula = CH



CAUSES AND REMOVAL OF IMPURITIES FROM PRECIPITATE

The greatest source of impurities is the result of chemical and physical interactions occurring at the precipitate's surface. The cations and anions at the precipitate's surface carry, respectively, a positive or a negative charge because they have incomplete coordination spheres.

In a precipitate of AgCl , for example, each silver ion in the precipitate's interior is bound to six chloride ions. A silver ion at the surface, however, is bound to no more than five chloride ions and carries a partial charge.

The presence of these partial charges makes the precipitate's surface an active site for the chemical and physical interactions that produce impurities.

Common Impurities

Common Impurities

The common impurities include;

(b) Inclusion

This is as a result of a potential interfering ion whose size and charge is similar to a lattice ion, may substitute into the crystal lattice structure. The probability of forming an inclusion is greatest when the concentration of the interfering ion is substantially greater than the lattice ion's concentration. An inclusion does not decrease the amount of analyte that precipitates provided that the precipitant is present in sufficient excess.

An inclusion is difficult to remove since it is chemically part of the precipitate's lattice. The only way to remove an inclusion is through **REPRECIPITATION**.

— Reprecipitation involves the isolation of the precipitate from the supernatant solution, dissolving it by heating in a portion of a suitable solvent. When then allow the solid to cool, reforming the precipitate. We can repeat the process of reprecipitation until the inclusion's mass is insignificant.

2. OCCULSIONS :

Occlusion form when interfering ions become trapped within the growing precipitate. Occlusion is localized, either along planes within the precipitate lattice structure or within aggregates of individual precipitated particles. An occlusion usually increases a precipitate's mass.

We can minimize occlusions by maintaining the precipitate in equilibrium with the supernatant solution for an extended time. This process is called a DIGESTION. During digestion, the dynamic nature of the solubility equilibrium, in which the precipitate dissolves and reforms, ensures that the occlusion is exposed to the supernatant solution. Because the rates of dissolution and reprecipitation are slow, there is less opportunity for forming new occlusions.

3 After Precipitation is complete, the surface continues to attract ions from solution. These SURFACE ADSORBS comprise a third type of impurity.

We can minimize surface adsorption by decreasing the precipitate's available surface area. We can also remove surface adsorbates by washing the precipitate. Although the potential loss of analyte cannot be ignored

CHM 221
Important points to remember
The number of molecules
such that they come
The molecular
placed in the
being de
12.5

CHM 221

Important points to remember in setting up gravimetric factors are:

The number of molecules of each of the two substances in a gravimetric ratio must be such that they contain the same number of atoms of the particular element involved.

The molecular weight (or a proper multiple) of the substance weighed (ppt) is always placed in the denominator, and the molecular weight of the sought substance (constituent being determined) is always placed in the numerator.

In setting up the factor, only the substance weighed and the substance sought are used.

The weight of the analyte is calculated from the amount of analyte contained in the precipitate by using the gravimetric factor, GF.

Gravimetric Factor (GF):

$$GF = \frac{g \text{ analyte}}{g \text{ precipitate}}$$

$$GF = \frac{AW \text{ analyte (g/mol)}}{(FW \text{ precipitate (g/mol)})} \times \frac{a(\text{moles analyte})}{b(\text{moles precipitate})}$$

$$GF = \frac{AW \text{ of Analyte}}{FW \text{ of Precipitate}}$$

For example, the gravimetric factor for Ni in Ni(HDMG)₂ is:

$$GF = \frac{AW \text{ of Ni}}{FW \text{ of Ni(HDMG)}_2} = \frac{58.71}{288.92} = 0.2032$$

Dimethylglyoxime (DMG)

$$\% \text{ analyte} = \frac{\text{weight analyte (g)}}{\text{Weight Sample (g)}} \times 100\%$$

Analyte is a substance whose chemical constituents are as needed.

weight sample (g)

$$\% (w/w) \text{ analyte (g)} = \frac{\text{wt ppt (g)} \times GF \times 100\%}{\text{wt sample}}$$

Assignment: When a sample of impure potassium chloride (0.4500g) was dissolved in water and treated with an excess of silver nitrate, 0.8402 g of silver chloride was precipitated. Calculate the percentage KCl in the original sample.

CONCEPT OF PRECIPITATE FORMATION

Precipitation is the formation of a solid in a solution during a chemical reaction. When the reaction occurs, the solid formed is called the **precipitate**, and the liquid remaining above the solid is called the **supernate**.

SUPERSATURATION AND NUCLEATION

Nucleation

Nucleation is a physical reaction which occurs when components in a solution start to precipitated. In a simple example of nucleation, supersaturated sugar water is used to make rock candy, with the sugar crystals nucleating and growing into crystals. Solubility refers to particle size larger than 0.01 mm. For particle size smaller than 0.01 mm the solubility increases greatly with the decreasing particle size. In gravimetric analysis large particle size precipitates. They are less prone to surface adsorption, easy to filter and washed free from impurities.

It is known that different experimental variables influence the particle size. Some of the important ones are:

- Precipitate solubility,
- Temperature,
- Reactant,
- Concentration and
- Rate at which they are mixed.

The net effect of these variables can at least be qualitatively accounted for by a single property of the system known as relative supersaturation.

Supersaturation

A supersaturated solution is one which contains greater concentration of the solute than permitted by the equilibrium solubility at the temperature under consideration. Essentially supersaturation is an unstable state just before the precipitation begins. This unstable state may be brought to a state of equilibrium by addition of a crystal of the solute or of some other substance or by mechanical means namely shaking or stirring.

Precipitate Formation

When mixing chemicals, it is often useful to know if a precipitate will form. This is done by computing an ion product P , which is analogous to the reaction quotient Q . P has the same form as solubility product K_{sp} and determines if a given solution will precipitate or not. A precipitate will form only if the instantaneous concentrations of the ions upon mixing exceed the solubility product of the insoluble substance.

If $P > K_{sp}$, the ion concentration is higher than the solution can absorb and the ions will form a precipitate.

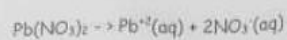
If $P < K_{sp}$, the ion concentration is lower than the solution can absorb and the ions not will form a precipitate.

If $P = K_{sp}$, the system is exactly saturated with ions and any extra amount of ion added will cause precipitation.

One thing to be careful of when computing the ion product is that you are most likely mixing two solutions. This will change the total volume of the mixture and thus all the concentrations will change.

Example: If a solution of 100 ml of 0.010 M $Pb(NO_3)_2$ and 100 ml of 0.010 M KF are mixed together. Will a precipitate of PbF_2 form? K_{sp} for $PbF_2 = 7.1 \times 10^{-7}$?

Example: We need to compute P and compare it to K_{sp} . We must be careful though- the solution volume changes when we mix the solutions, so we must adjust the concentrations of both ions. First, write out the ionization expressions

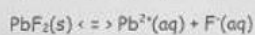


We start with 100 ml of 0.010 M lead nitrate. This means we have 0.0010 moles of lead nitrate, which forms 0.0010 moles of Pb^{2+} . We also have 100 ml of 0.010 M KF, 0.0010 moles of KF, which generates 0.0010 moles of F^- . Our total solution volume is 100ml+100ml = 200 ml, so the concentrations of the species are

$$0.0010 \text{ mol } \text{Pb}^{2+} / 0.200 \text{ L} = 0.0050 \text{ M } \text{Pb}^{2+}$$

$$0.0010 \text{ mol } \text{F}^- / 0.200 \text{ L} = 0.0050 \text{ M } \text{F}^-$$

Now we can compute P . It has the same form as K_{sp} .



$$P = [\text{Pb}^{2+}][\text{F}^-]^2$$

Simply plug in the needed values and compare to K_{sp} .

$$P = [\text{Pb}^{2+}][\text{F}^-]^2 = (0.0050)(0.0050)^2$$

$$P = 1.3 \times 10^{-7}$$

$P < K_{sp}$, so the mixture does not precipitate

$$K_{sp} \text{ for } \text{AgCl} = 1.0 \times 10^{-6}$$

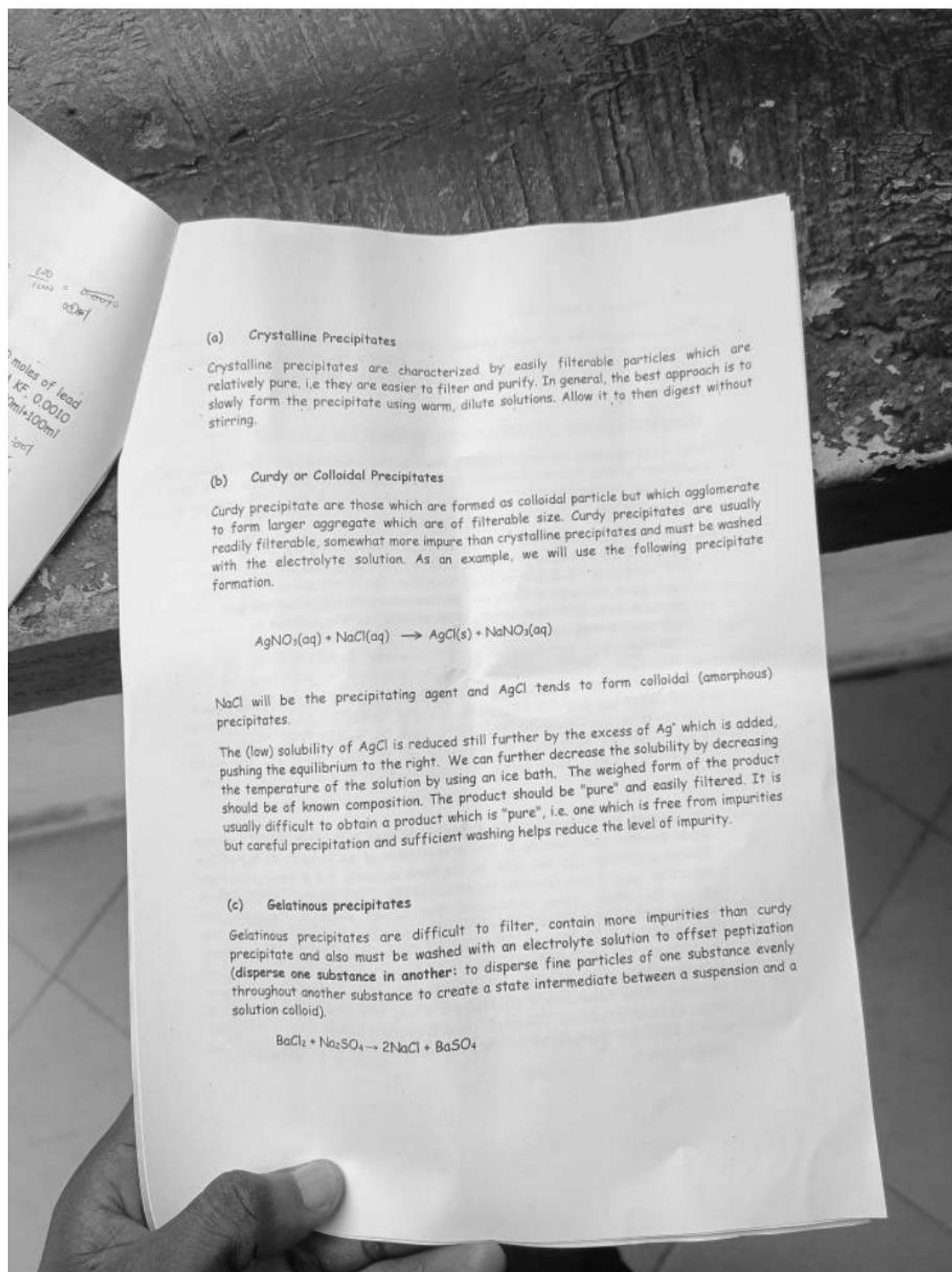
$$K_{sp} \text{ for } \text{PbSO}_4 = 1.6 \times 10^{-8}$$

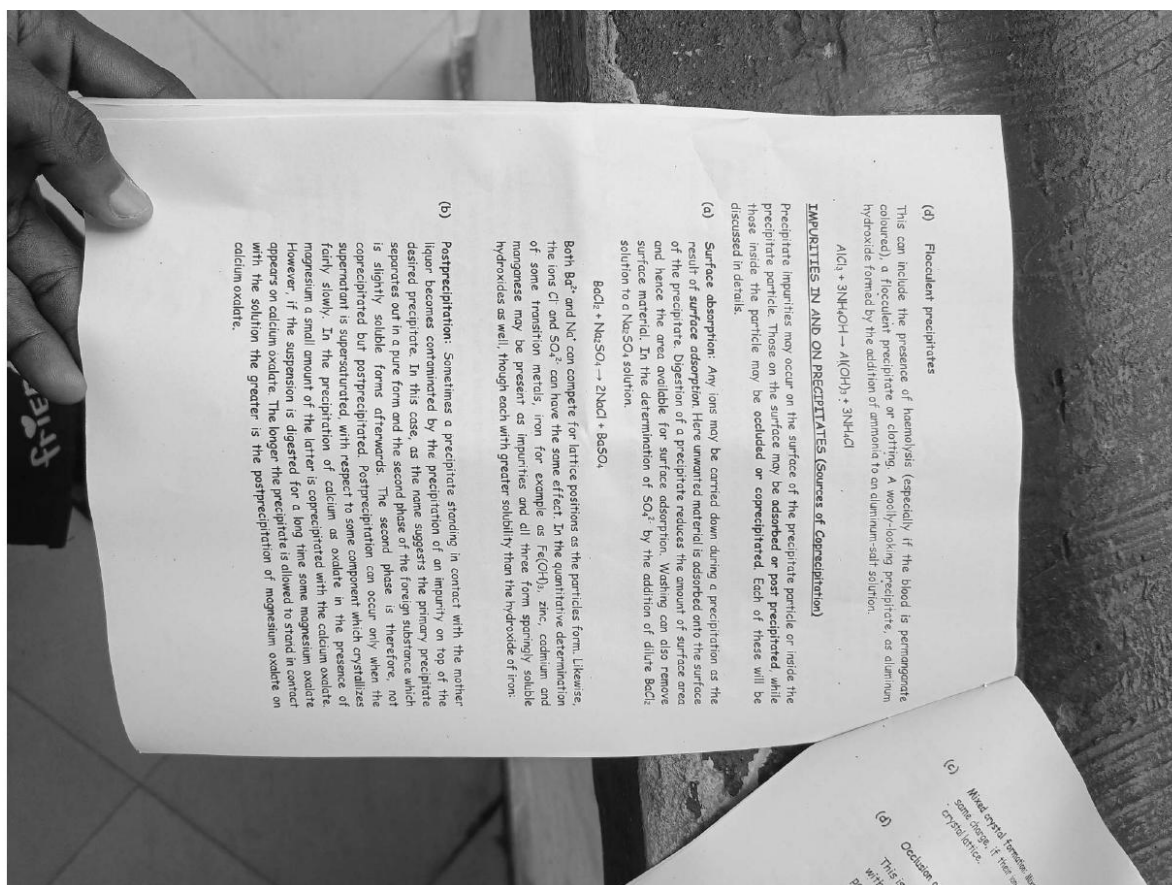
$$K_{sp} \text{ for } \text{AgBr} = 4.0 \times 10^{-13}$$

TYPES OF PRECIPITATES

Precipitates are classified as crystalline, curdy or colloidal, gelatinous and flocculent.

(a) Crystalline
Crystalline
relative
slowly
stir





(c) **Mixed crystal formation:** *Mixed crystal formation* can occur if two ions have the same charge, if their ionic diameters are sufficiently close to fit into the same crystal lattice.

(d) **Occlusion and Mechanical entrapment:**

This is a type of coprecipitation in which impurities (contaminants) are trapped within the growing crystal. If a precipitation procedure is carried out too quickly, pockets of solvent and spectator ions can form trapping them within the precipitate particles.

(e) **Coprecipitation**

Coprecipitation is the process of otherwise soluble material removed from solution during precipitate formation. Precipitates separating from a solution are not as pure but contain larger amount of foreign substances including mother liquor. The contamination of the precipitate is called coprecipitation. A typical example of conditions of precipitation is encountered when barium chloride is added to a solution of potassium sulphate the precipitated barium sulphate is found, after washing, to contain greater or lesser amounts of potassium sulphate inspite of the fact that this salt is freely soluble under the circumstances. Coprecipitation prominently takes place by the two mechanism i.e.

adsorption at the surface of the particles

occlusion by incorporation of foreign ions (and solvent)

PURIFICATION OF PRECIPITATES

Precipitate impurities must often be removed or reduced in order to obtain a satisfactory result in a determination. Some impurities may be removed by simple washing but most require more sophisticated treatment. Adsorbed impurities may be removed by simple aqueous washing of crystalline precipitates but the wash solution must contain an electrolyte to offset peptisation for curdy or flocculent precipitates. In this case, the electrolyte ion actually replaces one of the adsorbed ions, either in the primary or the secondary layer. If the precipitate is to be dried, the compound formed and left on the precipitate must be volatile.

Post-precipitate impurities cannot be removed by simple washing either with or without an electrolyte. Post-precipitate impurities may be reduced by dissolving the precipitate and reprecipitating being careful to filter as promptly as possible after the precipitate is formed.

Occluded and coprecipitation impurities are not affected by washing since they occur in the body of the precipitate particles. In some cases, they can be removed only by solution and reprecipitation. The amount of impurities occluded or co-precipitated will be less in the second precipitation since their concentration will be lower.

Some precipitates may be partially purified by aging or digestion. Aging is allowing the precipitate to remain in contact with the mother liquor for extended periods of time, and digestion is the same process except at elevated temperatures for shorter lengths of time.

Very small particles of the precipitate are more soluble than larger particles. Since the mother liquor constitutes a saturated solution, small particles will tend to dissolve during aging or digestion and deposit on the larger crystals. Thus, the small particles will disappear and the larger particles will grow larger. The impurities trapped during the growth of the smaller particles will be returned to the solution and probably will not be trapped during the slow growth of the larger particles. Even though the particles are not small enough to have increased solubility, the smaller particles will dissolve during aging or digestion and reprecipitate on the larger particles since the smaller surface area represent a more stable energy relationship.

WAYS TO ELIMINATE CONTAMINATIONS IN PRECIPITATES

(1) WASHING AND FILTERING

Washing

When the precipitation is carried out from the solution invariably only one of the constituents is precipitated keeping many other in solution. The object of washing the precipitate is to remove the impurities contaminating the precipitate as completely as possible. In the process of washing only surface impurities will be removed. This means

(2) Digestion
Digestion or precipitate aging at a higher temperature.
• Helps eliminate chemical process.

• Digestion

• The

• T

(2) DIGESTION

Digestion, or precipitate ageing, happens when a freshly-formed precipitate is left, usually at a higher temperature, in the solution from which it is precipitated. The physico-chemical process underlying digestion is called Ostwald ripening.

- Helps eliminate colloidal suspensions which are difficult or impossible to filter
- Digestion occurs when a precipitate is allowed to stand in its heated solution
- The solid re-dissolves and precipitates, along with increased crystal growth
- The resulting solid is usually more filterable and purer and bigger
- Minimizes coprecipitation

(3) PRECIPITATE DRYING

After filtration, the precipitate must be dried to constant weight. Generally the solids are dried at about 120°C but conditions for drying can vary considerably. Then remove excess solvent and drive off any volatile species. In some cases, the precipitate is heated to a point where it decomposes to a stable form for weighing. To determine the correct drying regime, thermogravimetric balances can be used to determine optimum drying time and temperatures.

RULES FOR ACHIEVING GOOD PRECIPITATION

- Elevated Temperatures- enhance solubility of the precipitate
- Dilute solutions
- Slow addition of precipitating agent
- Good stirring in heated solutions
- Careful pH control (if solubility is dependent on pH)